

Electromagnetism -- Maxwell's Equations

Subject: Electromagnetism | Topic: Unified Vector Field Theory & Electromagnetic Waves

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1. Differential Maxwell Equations in Free Space

In vacuum without charges or conduction currents, the electromagnetic field obeys:

1. Gauss's Law for Electricity: $\text{div}(\mathbf{E}) = 0$
2. Gauss's Law for Magnetism: $\text{div}(\mathbf{B}) = 0$
3. Faraday's Law of Induction: $\text{curl}(\mathbf{E}) = -\frac{d(\mathbf{B})}{dt}$
4. Ampere-Maxwell Law: $\text{curl}(\mathbf{B}) = \mu_0 * \epsilon_0 * \frac{d(\mathbf{E})}{dt}$

2. Displacement Current

James Clerk Maxwell identified that Ampere's original circuit law violated charge conservation for non-steady currents (e.g. charging capacitors). Adding the displacement current term $\mu_0 * \epsilon_0 * \frac{d(\mathbf{E})}{dt}$ restored continuity and predicted wave propagation.

3. Derivation of the Electromagnetic Wave Equation

Taking the curl of Faraday's Law and utilizing the vector identity $\text{curl}(\text{curl}(\mathbf{E})) = \text{grad}(\text{div}(\mathbf{E})) - \nabla^2(\mathbf{E})$ uncouples the fields into the wave equation:
 $\nabla^2(\mathbf{E}) - \mu_0 * \epsilon_0 * \frac{d^2(\mathbf{E})}{dt^2} = 0.$

4. The Speed of Light

The wave speed $c = 1 / \sqrt{\mu_0 * \epsilon_0}$ matches the measured speed of light, demonstrating that light is an electromagnetic disturbance.